AI & ML: Brief History

CS60050: Machine Learning

Abir Das & Ayan Chaudhury

IIT Kharagpur

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Agenda

§ Walking down the lanes of AI and ML history

Resources

- Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig.
- NPTEL Lectures on An Introduction to Artificial Intelligence by Prof. Mausam, IIT Delhi.
- Stanford course on Artificial Intelligence: Principles and Techniques (CS221), Autumn 2025 by Prof. Percy Liang.
- The slides are primarily inspired by these resources if not shamelessly copied.

Turing Asks the Question

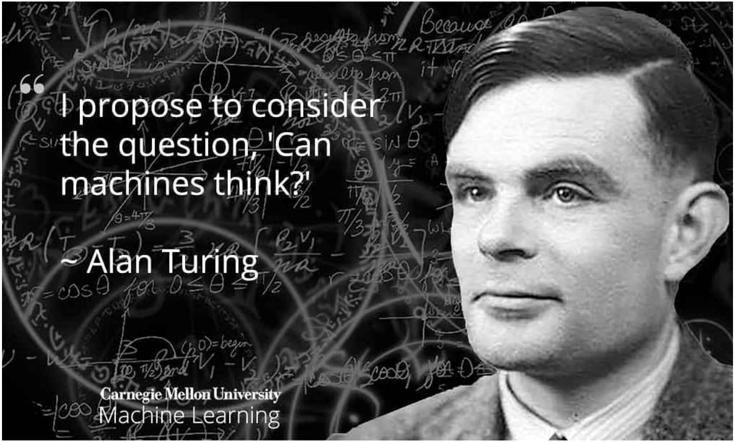

Image taken from: *CMU Machine Learning Department, FB Page*

- 1950: Alan Turing's landmark paper, Computing Machinery and Intelligence.
- This paper is remarkable not because it proposed any methods, but because it framed the philosophical discussions for decades to come.

The Turing Test

- A machine is said to pass the *Turing test* if it can convince a human judge that it's actually a human through natural language dialogue.
- Turing discusses two possible approaches.
  - The first is based on solving abstract problems like chess, which is the route taken by symbolic AI.
  - The second is where you build a machine and teach like a child, which is the route taken by neural and statistical AI.
- History shows symbolic, neural and statistical routes dominated the AI landscape in different eras, sometimes coming back to life from oblivion.

The Birth of AI

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- McCarthy convinced Minsky, Claude Shannon, and Nathaniel Rochester to help him bring together U.S. researchers interested in the study of intelligence.
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- The proposal mentioned: *An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.*
- The Dartmouth workshop did not lead to any new breakthroughs, but it did introduce all the major figures to each other and promoted why it was necessary for AI to become a separate field.

Early Successes & Overwhelming Optimism

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Underwhelming Results!

The famous retranslation of "the spirit is willing but the flesh is weak" as "the vodka is good but the meat is rotten" illustrates the difficulties encountered.

- 1966: ALPAC report cut off government funding for Machine Translation, first AI winter.

Knowledge-based systems (70-80s)

Expert systems: elicit specific domain knowledge from experts in form of rules:

Example Rule

IF: 1) The stain of the organism is gramneg, and
2) The morphology of the organism is rod, and
3) The aerobicity of the organism is aerobic
THEN: There is strongly suggestive evidence that the class of the organism is enterobacteriaceae

Knowledge-based systems

Wins:

- § Knowledge helped both the information and computation gap
- § First real application that impacted industry

Problems:

- § Deterministic rules couldn't handle the uncertainty of the real world
- § Rules quickly became too complex to create and maintain

"In order to grasp any part, it is necessary to understand how it fits with other parts, presents a dense mass, with no easy footholds. Even having written the program, I find it near the limit of what I can keep in mind at once."
— *Terry Winograd*

1987: Collapse of Lisp machines and second AI winter

Artificial neural networks

§ Early history:

- **1943:** artificial neural networks (McCulloch/Pitts)
- **1949:** "cells that fire together wire together" (Hebb)
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- Revival of connectionism:
  - **1980:** Neocognitron/CNNs (Fukushima)
  - **1986:** Backpropagation (Rumelhardt, Hinton, Williams)
  - **1989:** Handwritten digits for USPS (LeCun)

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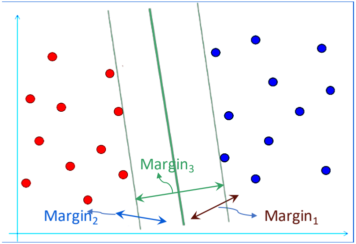
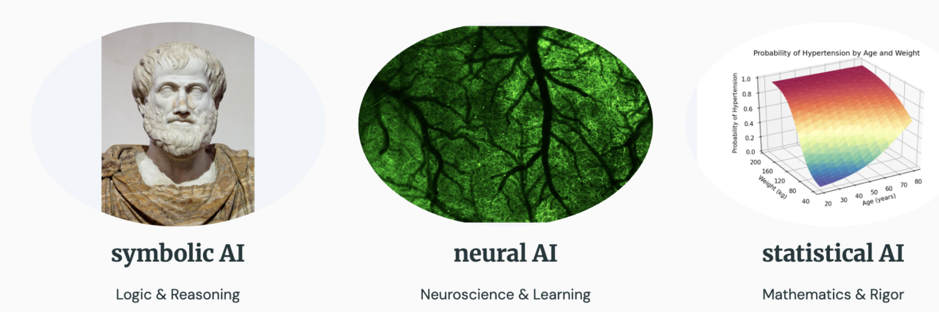
- § (2012): AlexNet obtains huge gains in object recognition; transformed computer vision community.
- § (2016): AlphaGo uses deep reinforcement learning, defeats world champion Lee Sedol in Go

Statistical machine learning

- In the middle, the field shifted focus toward mathematical rigor, temporarily moving away from "thinking machines" to "learning from data."
- **1985:** Bayesian networks (Pearl)
- **1995:** Support vector machines (Cortes/Vapnik)



Three Intellectual Traditions



§ "AI War" between these traditions are over. Modern AI is a **synthesis**.

Images taken from: *wikipedia*, *news.berkeley.edu*, *blog.stata.com*